

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2456653.90445 +/- 0.00049 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.186 +/- 0.013
Transit depth (Rp/Rs)^2	3.46 +/- 0.47 [%]
Orbital Inclination (inc)	80.54 +/- 0.85
Airmass coefficient 1 (a1)	1.184 +/- 0.012
Airmass coefficient 2 (a2)	-0.1137 +/- 0.0058
Scatter in the residuals of the lightcurve fit is	1.0 %
Best Comparison Star	None
Optimal Aperture	3.99
Optimal Annulus	11.6
Transit Duration (day)	0.0473 +/- 0.0036

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.186 $\pm$ 0.013 vs 0.1594 (deviation 2.0 σ)
Duration fit vs published	0.0473 d vs 0.0512 d (deviation 1.07 σ)
Mid-time O-C	1.6 min
Depth SNR	14.0
Residual scatter	1.0 %

Light curve

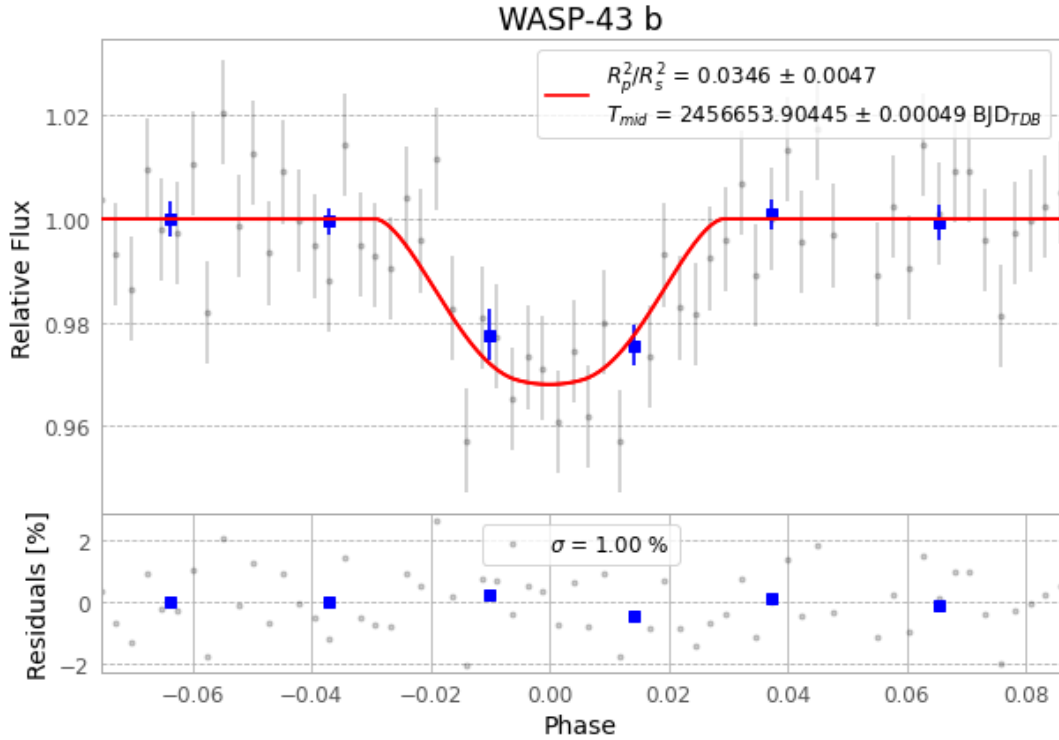


Figure 1 — FinalLightCurve_WASP-43 b_2013-12-27.png (SHA-256 555c80d120ce624f...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [181, 219], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 2ea378a5fe278b4f...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

61 FITS frames (40 MB), WASP-43131227080912.FITS → WASP-43131227111812.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-43-131227.log (7885a2b5c269...), FinalLightCurve_WASP-43 b_2013-12-27.png (555c80d120ce...), FinalLightCurve_WASP-43 b_2013-12-27.csv (fe10c1547734...)

Compute resources

Wall time 113.1 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-29 23:42Z.